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Friction and wear characteristics of chemically modified mahua (*madhuca indica*) oil based lubricant with SiO_2 nanoparticles as additives

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ABSTRACT

There is a tremendous increase in global warming due to the pollution enhanced by petroleum product utilization. Lubrication of the machinery parts is the main requirement for the industries, but conventional lubricants are having major problems related to their drainage. So, providing an eco-friendly oil and having better lubricity is a major task in recent research activities. This work performed tribological characteristics of epoxidized *Madhuca indica* oil while considering SiO_2 nanoparticles as additives. According to the evaluation, the *Madhuca indica* oil contains more unsaturated fatty acids by means of Gas Chromatography-Mass Spectrometry (GC-MS). The unsaturated fatty acid in the oil was chemically improved by the epoxidation technique to make it more acceptable, and varying quantities of silicon dioxide (SiO_2) were added. The thermo-physical properties of the material, such as kinematic viscosity, viscosity index, pour point, and flash point, as well as its tribological performance, were assessed. An optical microscope was utilized to examine the worn surfaces. The tribological characterization was carried out on a pin on disc wear testing machine. At 100 °C, nanoparticles addition to chemically treated oil results rise in kinematic viscosity of 1.2 wt percent. The addition of SiO_2 nanoparticles resulted in a flash point rise, but the pour point decreased. The presence of SiO_2 in the epoxidized oil improves the consistency of distribution. The addition of 0.8 wt%SiO_2 nanoparticles reduced the coefficient of friction & wear rate. The improved surface wear was observed up to 0.8 wt% concentration of nanoparticles and the same has been verified by worn surface images.

1. Introduction

It is well understood that petroleum products pollute the environment, particularly the aquatic environment. Synthetic lubricants are utilized in petroleum products to reduce friction during the lubrication process [1–3]. Friction causes energy loss in practically all moving machinery, which can be mitigated to some extent by employing lubricants [4,5].

Mineral oil disposal is hazardous because it is not eco-friendly and non-biodegradable. As a result, it is vital to effort on bio-based lubricants that are equivalent to fossil fuel-based lubricants and satisfy the stipulated quality and applications [4,6,7]. Furthermore, features such as pour point (PP), viscosity index, and flash point (FP), variations in viscosity with temperature, as well as the associated mechanisms and their performance, must be understood [8,9]. Vegetable oil has specific properties like higher viscosity, high flash point, biodegradable, non-toxic, etc., that make them sustainable during tribological applications. However, the edible oil's thermal and oxidative stability is low, although it can be improved through chemical modification techniques such as transesterification, hydrogenation, epoxidation, etc [10–12].

There are various additives available in the market that improve lubricant properties. Anti-wear additives, extreme pressure additives, foaming agents, and detergents are the most commonly used additives.

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Nomenclature

GC-MS	Gas Chromatography-Mass Spectrometry
SiO_2	Silicon dioxide
PP	Pour point
FP	Flash point
MoS_2	Molybdenum disulphide
hBN	Hexagonal Boron Nitride
CuO	Copper oxide
ZnO	Zinc oxide
VI	Viscosity index
POD	Pin on disc
L:	Length in mm
D	Diameter in mm
mm	Millimeter
mL:	Milliliter
FTIR	Fourier transform infrared spectrometer
cm	Centimeter
rpm	Revolution per minute
ASTM	American Society for Testing and Materials
Wt.%	Weight percentage
XRD	X-Ray Diffraction
D	Size in nm
λ	Wavelength
d	Full width
Θ	Diffraction angle
μ_{ku}:K	inematic viscosity at 40 °C
μ_{kl}:K	inematic viscosity at 40 °C with VI of 0
HRC	Hardness Rockwell C
nm	Nanometer
N	Newt.on
S	Second
COF	Coefficient of Friction
H_2O_2	Hydrogen peroxide
μ_{kh}	kinematic viscosity at 40 °C with VI of 100
mg	Milli gram
KOH	Potassium hydroxide
g	gram
Kg	Kilogram
m	Meter
-OH-	Hydroxyl
UV-VIS	Ultraviolet–Visible Spectroscopy

Phosphorous, sulphur, and chlorine are anti-wear and high-pressure additives. They coat the metal surface with phosphates, sulphates, and chloride and eventually protect it from pressure. However, the mentioned additives are chemically poisonous. The nanoparticles perform the same function as additives and are environmentally friendly [8,9,13]. As per the literature review, nanoparticles employed as a lubricating oil additive can improve the tribological qualities of the base lubricant. However, the majority of the reported nano additives contain heavy metals, sulphur, or phosphorus as active atoms, which can pose environmental risks. Inorganic nanoparticles, on the other hand, agglomerate quickly in many media and have a poor dispersive ability in organic solvents and oil. As a result, many nanoparticle applications are extremely limited. However, several physical and chemical techniques can be used to solve the dispersion problem. Zhu et al. discovered that by employing hydrocarbons as surface modification agents in lubricating oils, SiO2 nanoparticles may be dispersed stably [14]. Many researchers have shown the improved wear behaviors with SiO2 nanoparticles as a lubricant additive [15,16].

Various investigations reported regarding nanoparticles for tribological applications deal with the concentration, shape, and size. However, these studies are mostly limited to synthetic lubricants. Zhang et al. [17] combined the ground nano molybdenum disulphide (MoS_2) with palm oil and tested it under four distinct conditions. With the inclusion of MoS_2 nanoparticles, the result demonstrates improved lubrication. The effect of SiO_2 and MoS_2 nanoparticles in combination with engine oil was studied by Xie et al. [18]. The inclusion of nanoparticles improves stress-bearing ability. When SiO_2 nanoparticles were introduced to desert date oil, Singh et al. [19] investigated the influence on tribological characteristics. An improvement was noted during interaction with the surfaces while in the process. The SiO_2 provides better results in reducing friction and wear of the surfaces when they come into contact with each other. According to Wang et al. [20], the tribological characteristics were improved by using hexagonal boron nitride (hBN) nanoparticles in castor oil. At low load and high speed, 5 wt% hBN nanoparticles performed well in terms of wear reduction. During working at a great load and short sliding rate, 1 wt% hBN nanoparticles reduced the coefficient of friction and wear. Metallic oxide particles, like hBN [21], CuO [22], MoS_2 [18], ZnO [8], and SiO_2 [15,18] have been studied by many researchers for their lubricating properties. Most investigations concluded that the proper particle weight percent is essential for improved lubricity. Non-edible vegetables are suitable as a feedstock for bio-based lubricants because they contain straight-chain fatty acids with double bonds between their structure, which may help resist applied pressure on the lubricant film. Also, non-edible oil are not included in the food cycle [23].

To the best of the author's best knowledge, no research on the application of SiO_2 nanoparticles to *Madhuca indica* oil has been done earlier. The *Madhuca indiaca* has been used as a lubricant in a few or limited research, but they don't have any significance related to the non-edible vegetable oils. Furthermore, the applicability of nanoparticle-suspended *Madhuca indica* oil as a bio-based lubricant requires more attention that imparts novelty in this work.

There are various non-edible vegetable seed oils available in India, such as Jatropha, Karanja, Polanga, Mahua (*Madhuca indica*), contributing around 25 million tons of oil. *Madhuca indica oil* is extracted from the kernel of the mahua tree (*Madhuca indica*), a medium to large tree that can be found all through India. It is a tropical plant that easily breeds and is widely distributed throughout Asia, including India, Malaysia, Taiwan, China, Thailand, and Singapore.

Madhuca indica is environment-friendly non-edible vegetable oil and can be used as a lubricant in small amounts. Hence, this work aims to utilize *Madhuca indica* oil, a bio-based lubricant, and analyze its sustainability with SiO_2 nanoparticles. To make the bio-based lubricant more suitable for the nanoparticles, a chemical modification process known as epoxidation was performed to reduce the size of the unsaturated fatty acids and improve their structure. Nanoparticle $SiO_2$is further mixed with the chemically modified to understand the changes in the lubrication process and their fundamental features such as pour point (PP), *VI* (viscosity index), viscosity, etc. A pin-on-disc (POD) tribometer was used to perform the tribological characterization. After tribological characterization, an optical microscopic study of the worn surface was conducted for checking the effect of nanoparticles on the surface during the contact.

2. Materials and methods

2.1. Characterization of fatty acid composition of oil

For the analysis, fresh *Madhuca indica oil* was procured from the local market of the city of Dehradun, India. The purchased oil was yellowish

in color. The triglycerides fatty acid in oil was calculated using the EN14103 European standard. A GC-MS (Gas chromatography-Mass spectroscopic analysis) (GC-10, Shimadzu, Japan) equipped with a capillary tube (30 mm L and 0.25 mm D) lined with a 0.25 m (Rtx-5 ms, Rextex) was used to evaluate the fatty acid composition. With a flow ratio of 24:1, reference oil was put into the split/column. The chosen carrier gas (Helium) flow rate was 2 ml/min. The temperature was kept at 250 °C for heating (customized to start at 120 °C, hold for 5 min, and then warm to 250 °C at a rate of 3 °C/min).

2.2. Chemical treatment of oil

The chemical treatment was executed to get better the oil's physicochemical properties. The epoxidation method was used for a fundamental change of the base lubricant. The process was performed in a three-necked round flask, as shown in Fig. 1. The temperature sensor was linked to the left side, while the middle neck was outfitted with a reflux condenser. The left behind neck was utilized for keeping the base oil and chemicals. Chemical reagents in the necessary amounts (250 ml fresh oil + 24 ml $C_2H_4O_3$ + 0.3 ml conc.-H_2SO_4 catalytic agent) were mixed to perform epoxidation. A magnetic stirrer machine was used to stir the prepared mixture fully. Simultaneously with the stirring procedure, an oil bath was maintained at a temperature of 0–10^{0}C. Exothermic reactions caused the temperature to rise above 65^{0}C, therefore the speed of the stirrer was set at 300 rpm. After the mixing process was completed, 30 wt% H_2O_2 (i.e. 206 ml) was poured dropwise for additional 30 min. After that, treatment was carried out at 500–700 rpm for 4 h. The mixture was left for 1 h before water was withdrawn from the flask's bottom. The mixture was again heated at 65°C to remove the remaining water in the solution. According to the ASTM standards, the final solution was further analyzed for physicochemical characteristics.

Nanoparticles have an important function in improving lubricity. The addition of nanoparticles causes the creation of a protective film at the moving border, which improves lubrication and lowers the asperities at the mating surfaces. The SiO_2 nanoparticles were chosen as an additive because of their polishing, mending effect, and higher dispersion ability in the base oil. Sigma-Aldrich, Bangalore, India, provided the SiO_2 nanoparticles with a size of 35 nm. The nanoparticles are dispersed in varying weight percentages in the epoxidized oil (0.4, 0.8, and 1.2). To improve dispensability, nanoparticles were blended with surface modifiers (Isopropyl alcohol) and added to chemically modified oil. A total of 50 ml of nano SiO_2 is used as a surface modification. A magnetic mixer was used to completely mix the solution for 30 min at a speed of 750 revolutions per minute. The nanoparticles are kept in a stable suspension by stirring. A 3-digit accurate weighing balance machine was used to determine the weight of the nanoparticles introduced to the isopropyl alcohol. An ultrasonicator (M/s Samarth Electronics, Thane) was used to agitate the produced mixture for 15 min at a frequency of 0.3 kW.

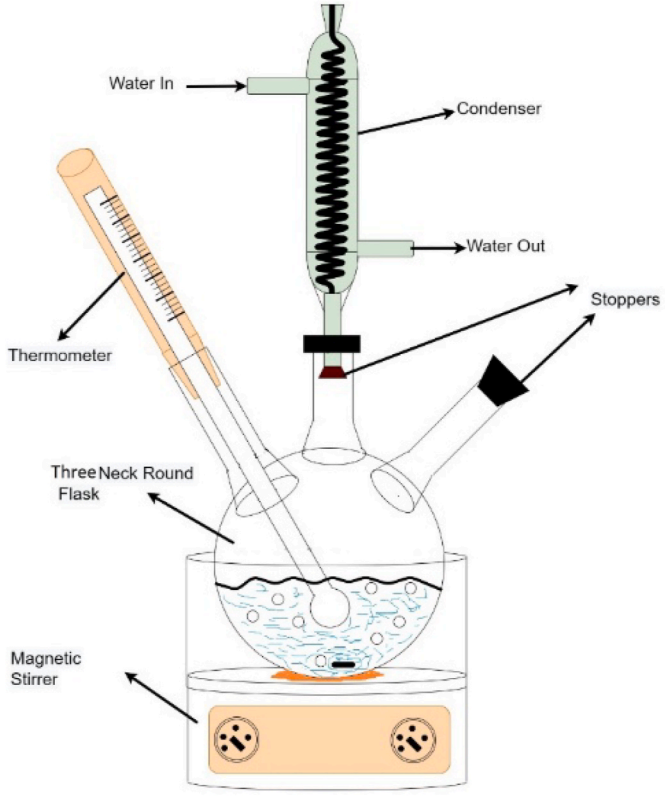


Fig. 1. Schematic details of 3-neck round-bottom flask.

2.3. XRD characterization

The nanoparticle structural investigations were carried out using XRD. Resulted peaks were recognized by Cu- Kα (λ = 1.5406 Å) with 40 kV and 30 mA with a 7°–80° (2α) angle range at a routing speed of 3° per minute. The size of the nanoparticles was determined using equation (1) (Debye-Scherrer) [22]:

$$D = \frac{0.9^{*}\ \lambda}{\mathrm{d}\cos\theta} \tag{1}$$

where:

D represents the average nanoparticle size in nm.Where, d and Θ are the widths at half of the maximum of the peak in radians and the peak diffraction angle, λ is the wavelength.

2.4. Physicochemical properties characterization

The kinematic viscosity was calculated using a viscometer (Make: Swastika SSL, Delhi). The test was carried out following ASTM standards for different properties. The data were taken at various temperatures (40^{0}C, 60^{0}C, 80^{0}C, and 100^{0}C) to check the details of lubricant film stability. For better results, the average of the three values was employed.

The viscosity index is calculated using equation (2), as shown below.

$$VI = \frac{[\mu_{kl} - \mu_{ku}]}{[\mu_{kl} - \mu_{kh}]} * 100 \tag{2}$$

Where:

μ_{ku} Denotes kinematic viscosity at 40°C, μ_{kl} and μ_{kh} signifies kinematic viscosity at 40°C with a viscosity index of 0 and 100, respectively.

The flash point (FP) and pour point (PP) were calculated per ASTM D-92 and ASTM D-97 standards, respectively. Whereas for calculating the acid number, Iodine number, and density, conventional procedures are followed as described earlier [24]. The physicochemical properties of the samples used in the analysis are listed in Table 1.

2.5. Fourier transformation infrared spectroscopy (FT-IR)

An FT-IR spectroscope examined the elimination of carbon bonds more unsaturated and the growing conformity of functional groups such as OH. The PerkinElmer 200i FT-IR Spectrometer system procured from M/s PerkinElmer Private Ltd., Thane, India, was used for the analysis by absorbing emissions obtained from the FT-IR. The test was accomplished with a resolution of 32 cm^{-1} utilizing a range of 4000–400 cm^{-1}.

2.6. Pin and disc material

The LM-13 alloy was purchased from the local supplier to make the

Table 1
Physicochemical properties of *Madhuca indica* and epoxidized oil-based lubricants.

Type of particle	Concentration (%)	Dimension (nm)	Density (kg/m^3)	Color	Shape
Silicon dioxide nanoparticle	99.90	35	2400	White	Spherical
Characteristics	***Madhuca indica oil***			**Epoxidized *Madhuca indica* oil**	
Kinematic Viscosity (mm^2/s) at 40°C	87.06			45.12	
Kinematic Viscosity (mm^2/s) at 100°C	11.03			7.12	
Density (kg/m^3) at 15°C	914.35			876.87	
FP (°C)	152.78			204.57	
PP (°C)	1.34			5.6	
Iodine value	19.06			8.87	
Acid value (mg KOH/g)	15.67			8.75	

pins. According to the literature, the hardness of the LM-13 alloy was 45 HRC [25]. LM-13 is a high-wear, high-corrosion-resistance aluminium alloy. EN 31 steel was used to make the disc. EN31 is exceptionally wearing resistant due to its high hardness value (65 HRC). The sample pins were produced in a cylindrical shape by using a lathe machine. The pin used had an 8 mm diameter and a 26 mm length. It was spherical from one side, and the other side was kept flat. The pin and disc were polished using emery papers with grit sizes ranging from 400 to 1200 nm.

2.7. Experimental set up

The NOVUS tribo solutions make machine was used to analyze the frictional behavior when the surfaces are in contact. The block diagram of pin and disc contacts is shown in Fig. 2. The friction coefficient was estimated using the frictional force and the applied load. Lubricating oil was pumped onto the disc at a 100 ml/min rate by an electric pump throughout the analysis. The worn surface analysis was done with the assistance of a Nikon Optical Microscope having a resolution in the range of 1000x to 2000x. The conditions applied during the test on a pin-on disc machine are mentioned in Table 2. The capacity of the device to apply load is limited to 200 N, so 150 N was selected for the analysis. According to the literature, the maximum wear occurred during low speed; therefore, 300 rpm was chosen for the examination [26–28]. Tribological characterization was accomplished by several laboratory-scale tribo-testers (such as pin-on-disk) under controlled settings with standard techniques, yielding higher accuracy [29–31]. These tribo-testers adhere to ASTM standards and provide effective testing procedures that are associated with real-world engine conditions.

2.8. Dispersion stability characterization

UV-VIS Spectrometer determines the dispersion stability of nano-particles and it was procured from M/s Lasany International, Panchkula, Haryana, India. The double beam spectrometer had a range of 190 nm–1100 nm. The beam was passed through the samples at different time intervals, and the absorbance was recorded for analysis.

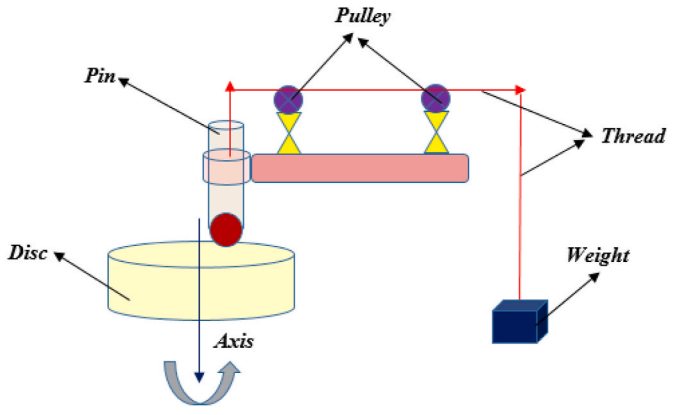


Fig. 2. Schematic image of the process during tribological analysis on a machine.

Table 2
Parameters for the test on the machine.

12	Data
Applied load (N)	150
Sliding speed (rpm)	300
Temperature (°C)	Room temperature
Duration (s)	3600

3. Results and discussion

3.1. Fatty acid composition characterization

Madhuca indica oil comprises 45.6% oleic acid and 18.7% linoleic acid, making up 64.3% unsaturated fatty acids with double and triple bonds. As a result, a chemical alteration was required to make it suitable for tribological applications. The number of chemical components found in *Madhuca indica* oil is shown in Fig. 3.

Mahua oil is characterized by free fatty acid (FFA) content of around 20% and a comparatively high percentage of saturated fatty acids such as stearic (14.0%) and palmitic (17.8%) acids. The remaining fatty acids are mostly spread among unsaturated components such as linoleic (17.9%) and oleic (46.3%) acids. The relatively high percentage of saturated fatty acids (35.8%) found in Mahua oil results in relatively poor low-temperature properties of the parallel methyl esters, as evidenced by PP value of 2 °C. So there are 64.3% of Saturated & 35.7% of

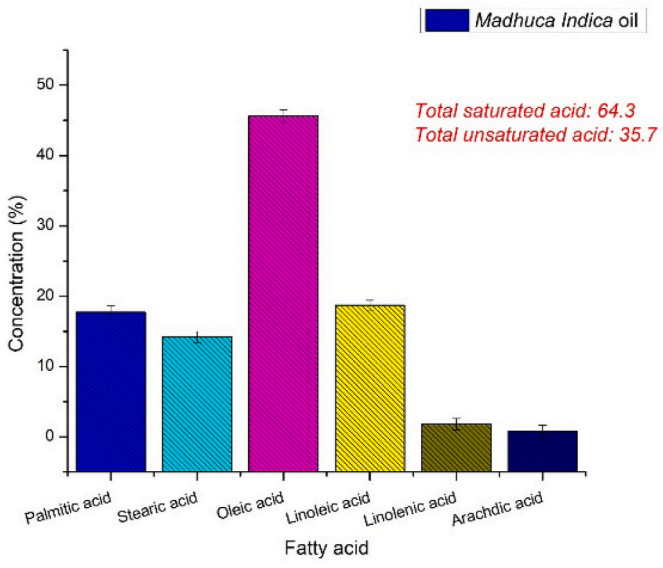


Fig. 3. Fatty acid chemical composition of *Madhuca indica* lubricant.

unsaturated fatty acid which makes it suitable for wear application and the effects of other chemicals are negligible.

3.2. Fourier transform infrared spectroscopy characterization

Fig. 4 Shows the FTIR results of the base oil (*Madhuca indica*) and modified *Madhuca indica* oil. Unsaturated fatty acids can be seen in the spectrum between 3000 and 3010 cm^{-1}, as illustrated in Fig. 4. At 1642 cm^{-1}, a lack of bonded carbon peaks was detected, clearly demonstrating that epoxidation has reduced unsaturated bonds. Fig. 5 shows the existence of C–O–C bonds at 827.29 cm^{-1}. The modified oil's FT-IR spectra were similar to the author's previously published research work [32]. In the 3225-3530 cm^{-1} spectral band, the presence of the –OH– group in the changed *Madhuca indica* lubricant was observed.

3.3. XRD analysis

By using equation (1) and Fig. 5, $\lambda = 1.5406$, $\Theta = 2^o$ *and* $d = 0.0346$ results in nanoparticles size of 36.02 nm. The size of the nanoparticles supplied by the vendor is validated by using equation (1) and Fig. 5. The crystallinity of the SiO_2 nanoparticles could be clear by sharp diffraction peaks obtained in Fig. 5. The diffraction peaks located at 12.23°, 17.45°, 21.34°, 39.23°, 54.12°, 66.34°, and 72.34° correspond to the (0 0 1), (0 0 2), (0 1 1), (0 1 2), (1 1 0), (0 1 3), and (0 2 0) reflection planes of tetragonal structure of SiO_2 (JCPDS no.33–1161, 85–1330 and 50–0927), respectively. The XRD pattern confirms that the nanoparticles have high purity as no peaks corresponding to impurities have been revealed. The SiO_2 nanoparticles computed unit cell parameters are a = b = 4.672 Å and c = 6.946 Å.

3.4. Dispersion stability of SiO_2 nanoparticles

The solubility of nanoparticles to the oil determines the dispersion stability. Nanoparticle solution was run via a UV-VIS spectrometer to study scattering stability. Fig. 6 indicates the dispersion ability of SiO_2 nanoparticles to the epoxidized *Madhuca indica* oil. An ultrasonic bath was given to nano lubricant samples at 25 °C for 45 min. The test was performed for 120 h for all the samples. No major changes were observed in the nanoparticle solution with a 0.4 and 0.8 wt% concentration. These conditions show that they got properly dispersed in the epoxidized oil. Some of the changes occurred in the 1.2 wt% solution due to precipitation that promotes their agglomeration at some particular points in the solution.

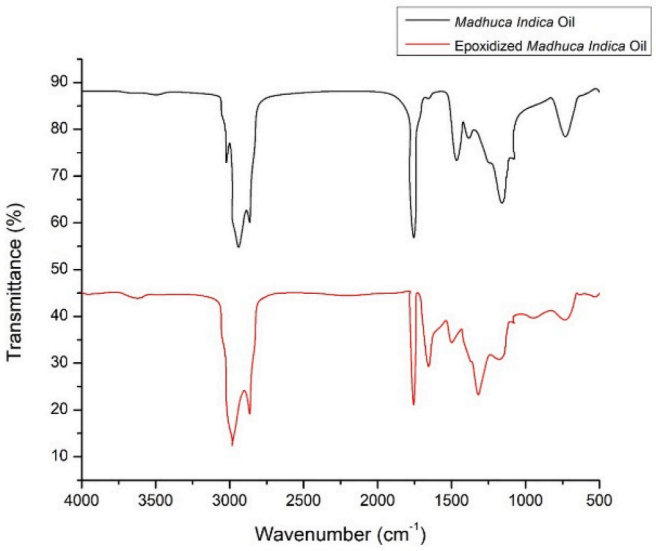


Fig. 4. FT-IR spectrum base *Madhuca indica* oil and epoxidized oil.

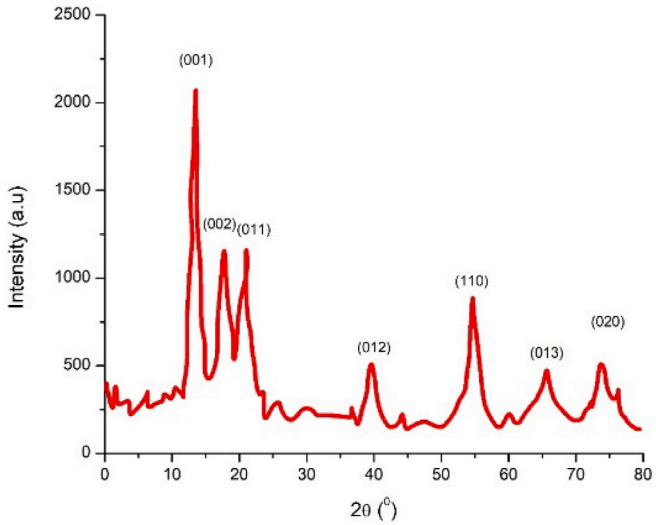


Fig. 5. XRD spectrum of SiO_2 nanoparticles.

3.5. Rheological properties characterization

3.5.1. Effect of temperature on viscosity

Fig. 7 depicts the change in viscosity as the amount of SiO_2 nanoparticles in the oil changes. In comparison to epoxidized oil, the addition of the nanoparticles enhances the thickness. The increase in temperature has been associated with a decrease in viscosity. There were not many changes in viscosity that occurred up to 0.8 wt% nanoparticles addition. With a 1.2 wt% increase, the greatest viscosity change was noted at 100^0C. The increased viscosity at a higher nanoparticle concentration was attributable to increased shear stress on the lubricant film surface. The more addition of nanoparticles after 0.8 wt% results in aggregation and micro collisions, which finish with increased thickness. The impact of the atomic forces increases the temperature, further resulting in lubricant viscosity decrement [22].

3.5.2. Increment in viscosity index (VI)

Another important component for lubricating characteristics is the viscosity index (VI), which is heavily impacted by kinematic viscosity (μ_k). The difference between VI with temperature and nanoparticle concentration is depicted in Fig. 8. With 1.2 wt% addition, the maximum viscosity was obtained at 100°C. There were slight changes observed in the epoxidized oil and nanoparticle concentration up to 0.8 wt%. The epoxidized oil has a straight-chain structure that helps retain its lubricant film concerning temperature changes. Up to a certain limit, the nanoparticles provide strength to the lubricant film that promotes better unevenness on the surfaces during their contact. At a greater temperature, complicated molecular interactions cause these changes.

3.5.3. Flash and pour point

The effects of SiO_2 nanoparticles on the flash point and pour point are depicted in Fig. 9. PP and FP increase as the weight percent of nano SiO_2 increases. The rise in flash point value provides a large temperature working limit, which delivers better strength and improves the thermophysical properties of nano SiO_2. As a result, an increase in the FP might be interpreted as a sign of better lubricity properties in modified oil. The presence of nano SiO_2 decreases the pour point. This occurred because of an enhancement in the physical and chemical properties of the lubricant-containing nanoparticles. The temperature rise adversely affects the nano lubricant movement and increases the weight percent of the nano additives doesn't deliver the improved movement [8,33–35].

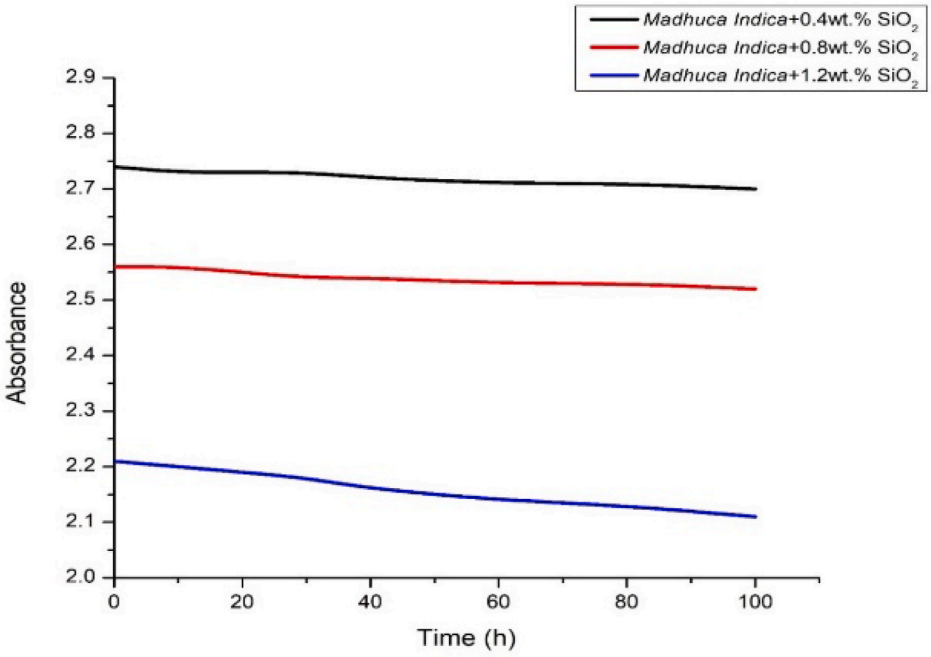


Fig. 6. UV-VIS Spectrometer analysis for the dispersion stability characterization of SiO_2 nanoparticles.

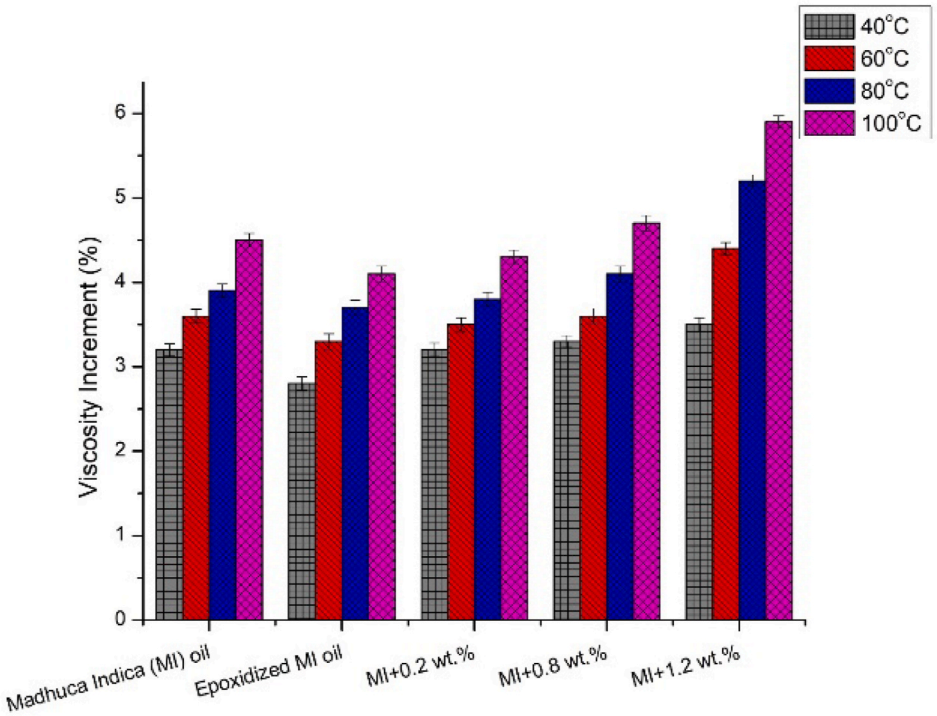


Fig. 7. Enhancement in viscosity percentage at different concentrations of oil.

3. 6. Coefficient of friction(COF)

As shown in Fig. 10, epoxidized *Madhuca indica* oil has the lowest COF compared to the base oil. A straight-chain complicated fatty acid structure was blamed for forming a thick film. The creation of an active film on the mating surface is seen when a saturated three-member ring forms while moving. Initial results show that base oil has the highest COF and takes longer to stabilize at contact surfaces.

Fig. 10 depicts the effect of nano SiO_2 on epoxidized oil. Because the film take time to grow on the face, changes in the coefficient of friction were noticed from 1750 s to 2000 s. At 0.4 and 0.6 wt% nano SiO_2, the least COF was recorded. When compared to the base oil, the decrease was roughly 2.34%. Concerning time, more stability of friction coefficient was observed with the introduction of nano SiO_2 particles as they form a stable layer on the surface quickly with the process.

The coefficient of friction in running conditions is lower for epoxidized oil than for base oil, and a steady-state condition is achieved. This could be owing to the enhanced tendency of epoxidized oil to become

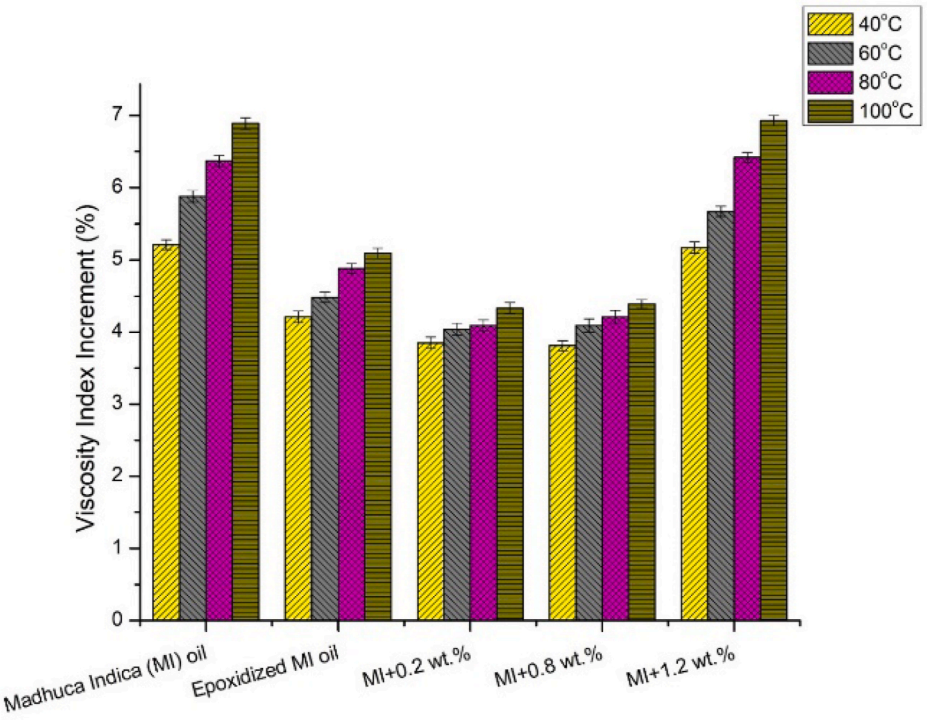


Fig. 8. VI variations with lubricant samples at different temperatures.

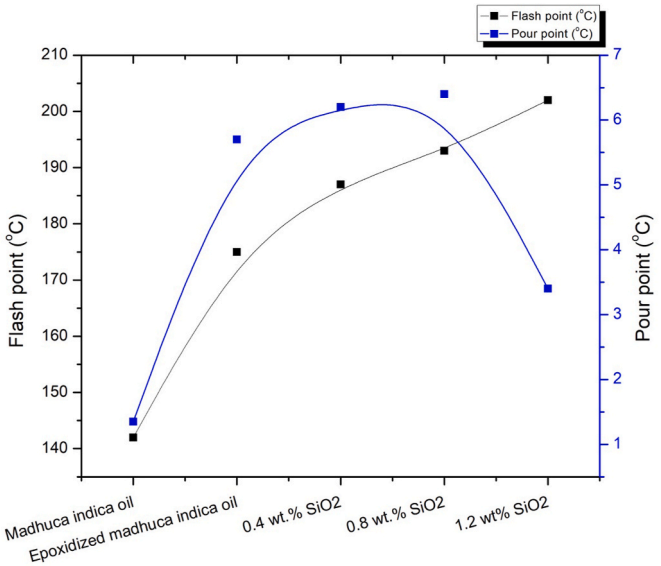


Fig. 9. Analysis of flash and pour point of the bio-based lubricants.

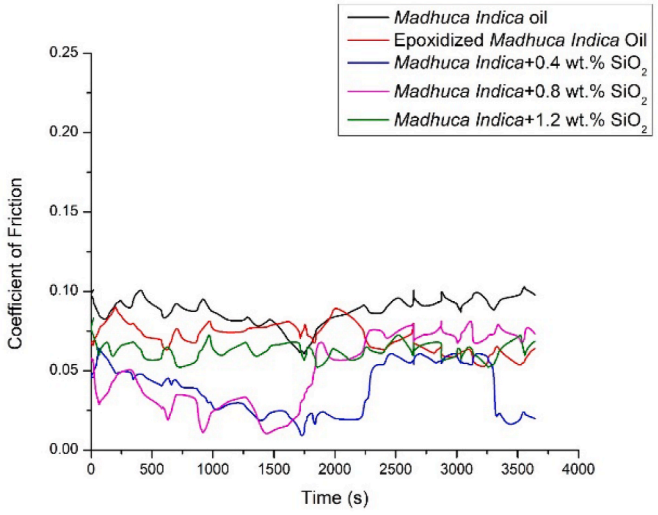


Fig. 10. Coefficient of friction with a base oil, epoxidized oil, and SiO_2 nanoparticles.

adsorbed on metal surfaces, avoiding direct metal-to-metal contact [36]. After the 1700s, COF increased due to a lack of effective layer development of lubricant breakdown, allowing direct metal-on-metal contact friction.

The mean COF of epoxidized oil containing 0.4 and 0.8 wt% SiO_2 nanoparticles is lower than that of epoxidized oil. This is because nanoparticles produce a strong tribofilm between moving surfaces, preventing direct metal-to-metal contact. The performance of 1.2 wt Percent SiO_2 nanoparticles differs significantly from that of 0.4 and 0.8 wt% concentrations. A more stable curve has been found, in contrast to the early reduction in coefficient of friction. This is owing to the sufficient and optimal solubility of SiO_2 nanoparticles in mahua oil, which results in the uniform development of SiO_2 films on sliding surfaces.

Estimating and developing a statistical correlation between engine performance metrics and tribological features of lubricants in real-world engines is difficult due to high modification costs and a huge number of lubricant samples. To avoid such complications, tribological characterization can be accomplished by several laboratory-scale tribo-testers (such as pin-on-disc) under controlled settings with standard techniques, yielding higher accuracy [29]. These tribo-testers adhere to ASTM standards and provide effective testing procedures that are associated with real-world engine conditions.

The smoother and greater lubricity fills holes with nano SiO_2 at the contacting surfaces. The use of nanoparticles enhances the contact area of the surfaces, resulting in the lower contact pressure and the occurrence of rolling rather than sliding phenomenon. The observed

explanation was found to be in good accord with Ghaednia et al. findings [37]. Fig. 11 shows the average coefficient of friction for the samples. It has been observed that significant changes also occurred at a low percentage in nanoparticle concentration.

Based on the results of the experiments, an optimal weight fraction of nano SiO_2 has been determined for establishing a better lubrication mechanism. There is no complete transition of the sliding action into a rolling motion at small concentrations, which creates minimal friction. Nano SiO_2 combines on the contacting surface at higher concentrations than a critical amount, resulting in increased rubbing and wear. When the concentration of nanoparticles is taken into account, the lubrication mechanism responsible is shown in Fig. 12. It is reasonable to assume that maintaining a suitable lubricating layer on the mating surfaces necessitates using the optimal concentration of nanoparticles. The surface repair is possible in the process where an optimum concentration of the lubricant is there, and it results in accumulation when they become more after an optimum limit. The rolling effect of nanoparticles is shown in the opposite direction in Fig. 12 as it depends on the surfaces in contact and they are opposite to each other. This shows the process shown by the nanoparticles during the motion within the lubricant film. Abrasive behavior is created by a higher amount of nanoparticles, decreasing nanoparticles due to the friction with the lubricant flow. Earlier investigations have stated a complete theoretical aspect that is comparable to this work [8,38,39].

3.7. Wear

Before the wear examination, the pin specimens were scrubbed with acetone & then dried out in an oven at a controlled temperature of 65°C. In comparison to the new lubricant, epoxidized madhuca *India* oil showed the least amount of wear Fig. 13. The epoxidized oil forms an enhanced lubricating film between the pieces in contact. The lubricant, after epoxidation, comprises polar molecules that increase the contacting surface adsorption capacity, resulting in less wear [40].

The addition of up to 0.8 wt% SiO_2 nanoparticles reduced sample wear compared to the new *Madhuca indica* lubricant. The reduction in wear was approximately more than the base oil. Wear resistance is improved by the homogeneous dispersion of nanoparticles of SiO_2 in oil. Singh et al. [25] investigated ZnO nanoparticles as additives, when compared to raw Euphorbia Lathyris oil, the concentration of ZnO nanoparticles as additives to this oil up to 0.5% likewise demonstrated minimal pin wear. The wear was reduced by approximately 6.73%. The

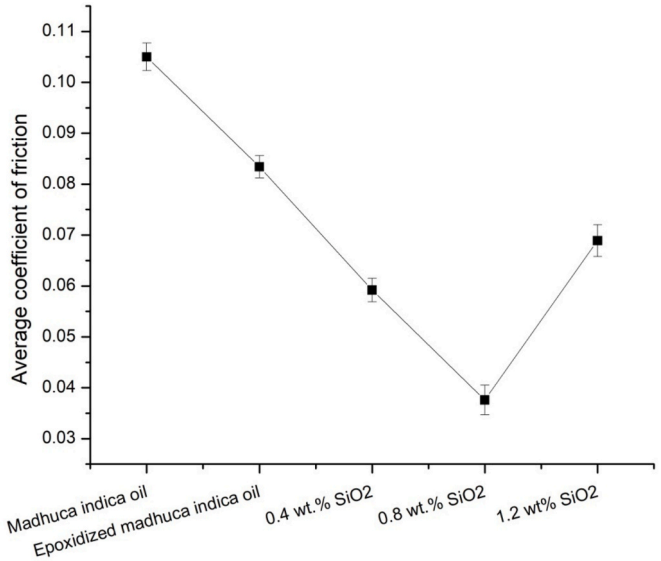


Fig. 11. Average coefficient of friction.

influence of nanoparticles on anti-wear properties is determined by particle dispersion in the lubricant. The nanoparticles have a superior ability to diffuse properly in the fluid, promoting the anti-wear mechanism.

3.8. Surface morphology

The optical microscopic pictures of the produced oil samples are shown in Fig. 14. The surface affected by base oil consideration is shown in Fig. 14 (a), and the surface has scratch marks. The disintegration of the layer on the surface caused delamination of the worn surface. A smooth surface was generated with the epoxidized *Madhuca indica* lubricant, as shown in Fig. 14 (b), and there were fewer scratches than with the base oil. The lubricating film is aided by creating -O- cross-linking on the surface, which preserves it [41–43]. The inclusion of nanoparticles up to 0.8 wt% provides reduces wear by forming a protective coating between the surfaces as they move. Images exhibiting the effect of 0.8 wt% SiO_2 nanoparticles are shown in Fig. 14 (c). The least amount of damage was detected, signifying a lessening in surface asperities due to proper nanoparticle loading and contact force resistance. However, more increase in nanoparticles causes an upsurge in wear. When the concentration of nano SiO_2 is increased further, the result of adhesion & abrasion on the worn faces is shown in Fig. 14 (d). This was attributed to the deposition of nano SiO_2 on contacting faces, which resulted in increased contacting face wear [35].

3.9. Lubrication mechanism

At room temperature, Fig. 12 schematically depicts the lubricating mechanism of the produced nano lubricants. The main lubrication process occur when SiO_2 nanoparticles are added to reference oil and results is the formation of the transfer layer due to relative sliding of the nanoparticles on both sides. The tribo-film generated and the practical SiO_2 nanoparticles injected into the contact zone have a major impact on the development of tribological features. Certain SiO_2 nanoparticles agglomerate as the concentration of SiO_2 nano lubricant is increased from 0.8 wt%, resulting in greater secondary particle size, as seen in Fig. 11. Friction and wear would be exacerbated, resulting in a higher COF and wear area. This means that the optimal concentration of SiO_2 nano lubricants is 0.8 wt%. The 0.8 wt% SiO_2 nano lubricant can deliver enough nanoparticles to the contact region on a continuous basis with minimal agglomeration. As a consequence, the optimal SiO_2 mass fraction is 0.8 wt%, which gives the highest tribological properties, such as the lowest COF and most effective wear resistance. Only a small number of SiO_2 nanoparticles are able to take effect in the SiO_2 nano lubricant with a concentration of less than 0.4 wt%, whereas the lubricant with a high SiO_2 concentration of 1.2 wt% permits SiO_2 nanoparticles to agglomerate. On the other hand, the agglomeration of SiO_2 nanoparticles acts as a barrier, restricting a constant supply of microscopic nanoparticles to the contact zone for lubrication. The pin and disc surfaces, on the other hand, would be severely abraded by the agglomerated nanoparticles. As a consequence, the lubricant with the best lubricating qualities was the one with 0.8 wt% SiO_2.

The production of a metallic lubricant film can be used to explain how silicon nanoparticles reduce friction and wear. The creation of the film can be aided by two methods. One by nanoparticles deposited on the worn surface, which was “fresh” by being removed surface oxide layer in sliding, through electrostatic attachment generated by sliding of the friction surfaces, according to one hypothesis based on chemistry and electrochemistry [44]. The chemical deposition of silicon together with a tiny amount of aluminium grindings on steel was then triggered by local overheating owing to direct contact of two surfaces [45], resulting in the formation of a metallic lubricant film containing a little amount of aluminium on the worn steel surface. The other alternative process can be explained using mechanical metallurgy principles: local overheating and high flash temperature on the friction surface despite

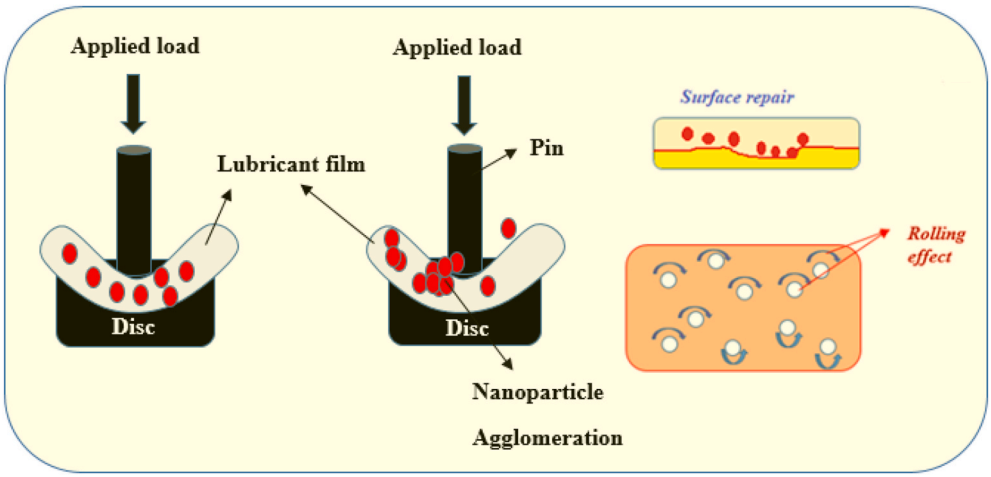


Fig. 12. Schematic image for the process of lubrication mechanism with the introduction of nanoparticles as additives.

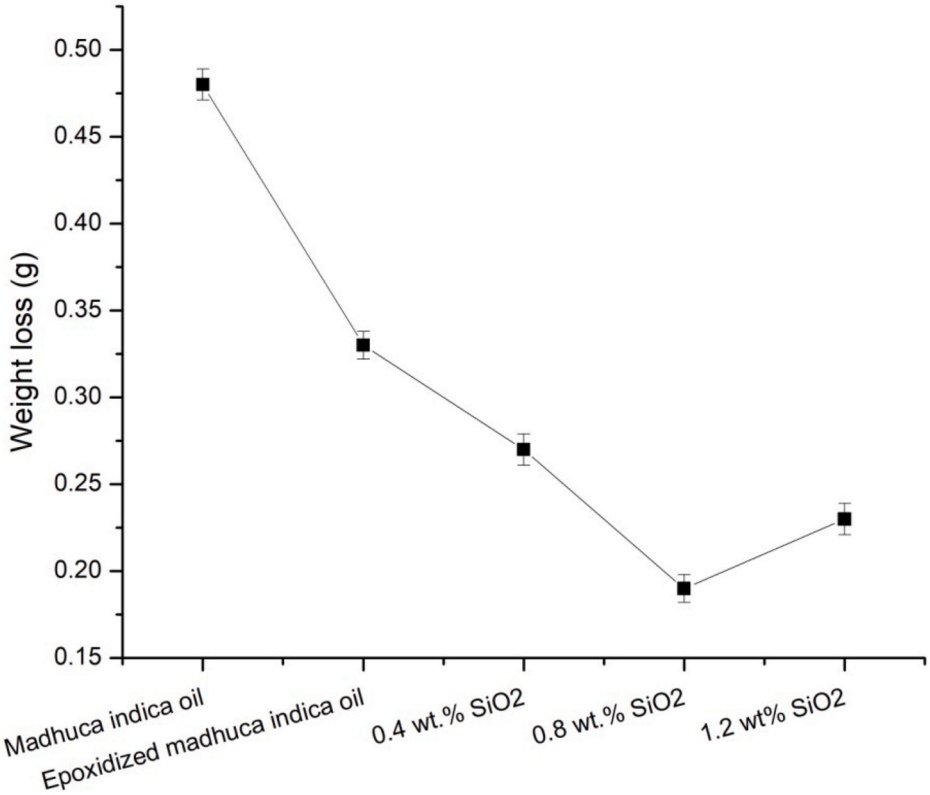


Fig. 13. wt loss of pin with lubricant samples.

oil cooling. On the steel surface, melting or semi-fused silicon has a good wetting property, and liquid silicon may well wet the "new" friction surface. The shearing movement can then smear it on the worn surface. As a result, a silicon protective coating may form as shown in Fig. 15 [30].

4. Conclusions and suggestions

The *Madhuca indica* lubricant was chemically processed, and SiO_2 nanoparticles were added at various weight fractions to the epoxidized lubricant, which was then tested for tribological characteristics. Based on the results of the experiments, enhanced behavior can be achieved by adding SiO_2 nanoparticles to the epoxidized lubricant. By introducing nanoparticles to the epoxidized oil, improvements and changes in lubricity have been observed. The following findings are based on the above experimentation:

- During the mixing of nano SiO_2 after a particular concentration, an increase in viscidity was seen during the analysis of physicochemical properties of the nano emollients relating to the nanoparticles and an augment in temperature.
- An increase in the VI percentage was seen when the nano SiO_2 was mixed. At a larger weight %, the rise in viscosity was observed.
- The FP of the epoxidized *Madhuca indica* lubricant is increased when nano SiO2 is added to the chemically modified lubricant. Compared to the epoxidized oil, the greatest flash point value was noted at 1.20 wt%, which was a 2.78% increase.

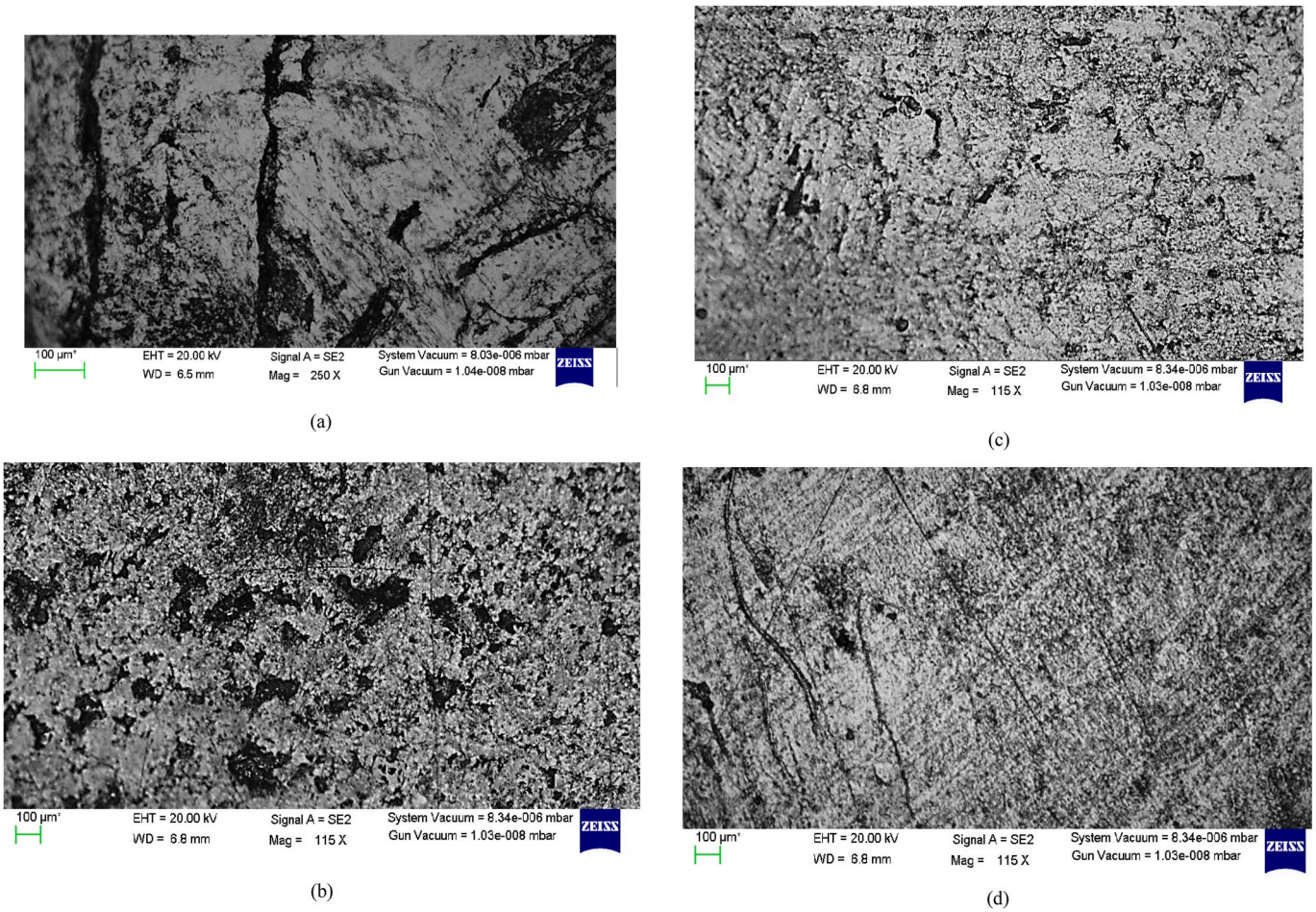


Fig. 14. Optical microscopic images of the worn surfaces for (a) *Madhuca indica* oil (b) epoxidized *Madhuca indica oil* (c) SiO_2 (0.8 wt%) (d) SiO_2 (1.2 wt%).

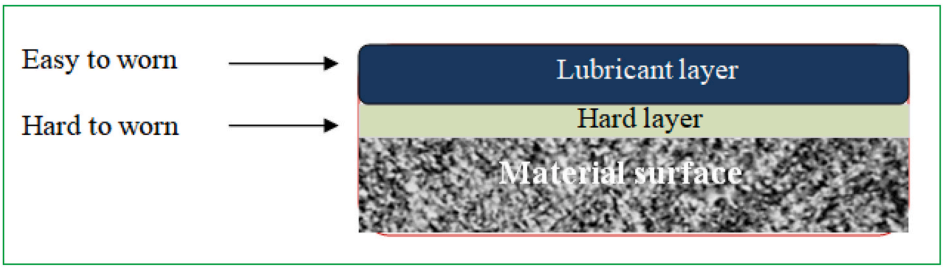


Fig. 15. Schematic image of the tribological worn process of the lubricant film.

- The PP is reduced by mixing nano SiO_2 into a clean lubricant. At lower SiO_2 weight percentages, such as 0.4 wt% and 0.8 wt%, there are only minor differences in the PP.
- When nano SiO_2 was investigated for a set period at room temperature, the proper scattering stability was discovered. The lowest precipitation was observed with maximum concentration of SiO_2, indicating that they are better distributed to the epoxidized lubricant.
- The *Madhuca indica,* after epoxidation, has shown improved lubricity and reduced friction compared to the base oil. This was attributed to -O- cross-linking on the contacting surfaces, successfully generating a protective film on the contacting faces.
- The addition of nano SiO_2 to an epoxidized lubricant improves its tribological properties. Improved results in reducing friction are reported at concentrations of up to 0.8 wt%. The addition of nano SiO_2 particles in chemically treated oil give to forming a greater protective barrier among the mating faces, preventing direct contact between moving surfaces.
- Compared to other oil samples, the base lubricant shows many marks, cuts and scratches due to the delamination and trident cuts phenomena on the worn surfaces. In distinction to the reference lubricant, the modified oil shows less wear on the haggard faces due to the presence of double and triple bonds present in the chemical structure. When nanoparticles are added to the epoxidized lubricant at a concentration of up to 0.8 wt%, the surface wear, and impairment are decreased. Because of restricted heating and melting of nanoparticles, the suitable adsorption of nano SiO_2 on the contacting surface occurred.

Nanolubricants are currently used in automotive applications such as engine and braking systems, as well as machining operations such as milling and grinding as a cutting fluid. Nanolubricants are used

extensively in automotive engines. The favorable effects of introducing nano-sized elements on the tribological performance of engine oils have attracted the interest of many researchers. It has been observed that adding nanoparticles to engine oils such as 10W40, SAE20W50, and SAE40 can significantly improve their lubricating effectiveness. The use of nanoparticle lubricants is another promising area for assessing the hydrodynamic performance of journal bearings. The use of nano-lubricants might be explored for evaluating the performance of worn-out journal bearings and bearings with higher surface roughness [46,47].

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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